Cerebral Palsy

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❖ **Overview**
- non-progressive neuro development disease
- Affecting mainly the motor centers
- Results in disorders of movement and posture

❖ **Associations**
- Mental retardation 50%
- Epilepsy 30%
- Impaired hearing ;deafness
- Impaired vision
- Emotional disturbances
- Behavioral disturbances

❖ **Classification**

➢ **Etiology**
- Prenatal
- Perinatal
- postnatal

➢ **Functional**

○ **Class 1**
- No limitation of activity

○ **Class 2**
- Slight limitation of activity

○ **Class 3**
- Moderate limitation of activity

○ **Class 4**
- No useful physical activity

➢ **Clinical**

○ **cerebral cortex [Spastic cerebral palsy] 70%**
- monoplegia
- Diplegia
- Paraplegia
- Hemiplegia
- Quadriplegia

○ Cerebellum [Ataxic cerebral palsy]
○ basal ganglia [dyskinetic cerebral palsy}
○ Mixed cerebral palsy
○ Hypotonic

❖ **Causes**

➢ **Pre-natal (80%)**
- Antenatal Infections
- Congenital malformations
- Fetal asphyxia
- cerebral infarction
- Metabolic & toxic factors

➢ **Natal (10%)**
- Birth asphyxia
- Birth trauma

➢ **Post-natal (10%)**
- VLBW with intracranial hemorrhage
- Meningitis, encephalitis
- Metabolic e.g. phenylketonuria
- Hypoglycemia
- Hyperbilirubinemia
- Hydrocephalus

❖ **Spastic cerebral palsy**

➢ **Criteria**
○ commonest type
○ **Pyramidal tract lesion (UMNL) signs:**
- Hypertonia
- Hyperreflexia
- Positive Babinski sign
- May be clonus
- Persistence of primitive reflexes
- Pseudobulbar palsy
  - feeding disorder (poor suckling & swallowing)
  - squint and speech disorders

➢ **Spastic diplegia**
○ Crawling is commando
○ Lower limbs scissoring
○ MRI typically shows scarring and shrinkage in the periventricular white matter with compensatory enlargement of the cerebral ventricles

➢ **Spastic hemiplegia**
○ Due to in utero or neonatal stroke
○ hand preference at a very early age
○ tiptoe walking
○ Circumdactive gait

➢ **Spastic quadriplegia**
○ most severe type
○ ischemia and infection
○ associated with mental retardation and seizures

❖ **Ataxic cerebral palsy**

➢ **Criteria**
○ Hypotonia and hyporeflexia
○ **Cerebellar ataxia**
- incoordination of voluntary movements
- nystagmus
- staccato speech
- intention tremors.

❖ **dyskinetic cerebral palsy**

➢ **Commonest causes**
○ **kernicterus**
○ Asphyxia

- **Criteria**
  - Hypotonia (replaced with time with hypertonia & rigidity)
  - Chorio-asthetoid movements.
  - Deafness
- **Diagnosis**
- **Hx & Ex**
  - **to rule out a progressive disorder of the CNS**
    - degenerative diseases
    - metabolic disorders
    - spinal cord tumor
    - muscular dystrophy
  - **Screening for comorbid conditions**
    - visual and hearing impairments
    - speech and language disorders
  - classify the type of cerebral palsy
- **CT & MRI**
  - rule out brain tumors & degenerative brain disease
  - detect the cause e.g. brain malformations and spinal cord lesions
  - detect calcifications associated with congenital infections
  - TORCH screen
  - Metabolic screen
  - Genetic evaluation
  - **For associations**
    - evoked potential test for Hearing Visual
    - EEG for seizures
- **DDx of cerebral palsy**
  - Spinal cord tumors
  - Channelopathies
  - Sandifer syndrome
  - MECP2 duplication
  - Congenital dopa-responsive disorders
  - Genetic spastic paraplegia
  - Some metabolic conditions (GLUTI deficiency, glutaric aciduria type I)
  - Ataxia telangiectasia
- **Management**
- **Objectives**
  - improve function
  - prevent deformity
  - encourage independence.
- **Multidisciplinary approach**
  - Occupational and physical therapists
  - Speech pathologists
  - Social workers
  - Educators
  - Developmental psychologists
- **Assist:**
  - Feeding & defecation
  - Vision and hearing
  - Walking
  - Communication
  - Rehabilitation of motor disability
- **Medications**
  - Anti spastic drugs
    - Diazepam
    - Baclofen
    - Dantrolene
  - Botox A
  - Levodopa
- **Surgery**
  - For marked spasticity to reduce muscle spasm
  - rhizotomy
  - adductor tenotomy
- **Early Signs of CP**
  - early head support (spasticity)
  - Moving one part
  - Persistent clenched fist
  - persistent neonatal reflexes
- **Notes**
  - half of all the cases of cerebral palsy are associated with preterm delivery and low birth weight
  - main pathological lesions found in preterm infants are periventricular leukomalacia and intracerebral haemorrhage